Subject: DATABASE MANAGEMENT SYSTEMS | **Type:** Lecture Notes & Study Material

Student: Kamichetty Veera Pradeepthi (23A91A1225) | **Source:** PEC.UP Smart Academic Portal

Unit 1: ER Model & Relational Algebra - Quick Revision Notes

1. Introduction to Database Management Systems (DBMS)

A DBMS is a collection of interrelated data and a set of programs to access those data. The primary goal of a DBMS is to provide a way to store and retrieve database information that is both convenient and efficient. Key characteristics include data independence, reduced data redundancy, transaction consistency (ACID), and robust security mechanisms.

2. Entity-Relationship (ER) Model

The ER data model employs three basic concepts: Entities (real-world objects distinguishable from other objects), Entity Sets (collections of entities of the same type), Attributes (properties describing entities), and Relationships (associations among several entities). Cardinalities include 1:1, 1:N, N:1, and M:N.

3. Relational Model & Fundamental Relational Algebra Operations

Fundamental operations: Selection (sigma) selects tuples satisfying a given predicate. Projection (pi) outputs specified attributes. Cartesian Product (cross) combines information from two relations. Set Difference (-) finds tuples in one relation but not another. Union (cup) combines tuples from two union-compatible relations.

Downloaded from PEC.UP Academic Resource Hub (https://pecup.in) • Internal Academic Circulation Only